

TOTALIZERS VERSUS CRITICS

What are the limits of knowledge? Throughout the history of human thinking this question has been raised again and again, and those who have been concerned with confronting it fall, roughly speaking, into two separate camps. First, there are those who claim that there are no inherent limits to human knowledge. The Presocratics, Plato, Aristotle, St. Anselm, the Epicureans, the Stoics, and Descartes might be classified, more or less, as representatives from this first camp. Their optimistic and ambitious attitude toward human learning includes a vision of totality, and they believe that through its powers the human mind is capable of unlimited and complete knowledge of the cosmos and its first principles. The human mind, according to this group of thinkers, is like a mirror that, when properly polished, reflects back an accurate and dependable image of the world in which we live. These are the metaphysicians, the systematizers, and the “big thinkers” whose philosophical speculations steer a course toward certainty. They are the ones who believe that philosophical thinking is capable of uncovering the vast entirety that is the universe and of making it wholly present in our thinking once and for all, thereby overcoming the need for any further philosophizing.

In the second camp are those who claim that human knowledge is inherently limited, and that no matter how far we push our inquiries we will always fall short of a total and final comprehension of reality. At their most philosophical moments, all of the individuals we have so far surveyed express this sentiment to one degree or another; however, it is in the thoughts of Socrates, Diogenes, the Skeptics, St. Augustine, St. Aquinas, and Hume that we have seen this attitude highlighted most vividly. These are the individuals whose major purpose is to humble us in the face of the infinitely unknowable, whether that be the universe, God, or ourselves. Human knowledge, they emphasize, never quite encapsulates the whole Truth, and so we should always remain modest about our intellectual claims. These sorts of thinkers have, from time to time, been accused of pessimism (and even nihilism) insofar as at least part of their philosophical projects are focused on demonstrating the shortcomings, and in fact the vanity, that is involved in the drive for absolute knowledge. There is always a facet of Being that eludes our intellectual grasp, according to these thinkers, and we are most wise when we are fully aware of, and willing to own up to, our ignorance of the Absolute.

This tension between what I will call the “totalizers” and the “critics” has, I believe, always been a part of philosophy. On the one hand, the totalizers represent that aspect of philosophy that strives for knowledge of absolute Truth and desires nothing more than its fulfillment. However, in their desire for the Absolute there is, perhaps, always a danger that these sorts of individuals will jump too quickly for the “golden ring” and thus come away with something less than the grand prize for which they had hoped. I think this is what many of us find awe-inspiring, yet also frustrating, in the totalizing visions of Plato, Aristotle, Descartes, etc. They aspire to so much, yet in the end their final systematic philosophies always deliver less than we hope for.

This is precisely why the critics, like Socrates and Hume, can be so delightful. They represent that aspect of philosophy that is always attuned to our separation from absolute Truth. They remind us that we are not gods, but humans with finite intellects who must digest the world one bite at a time. It is satisfying, for many of us, to see arrogant totalizers brought down to earth and be reminded of their failures by these critics. However, just as there is the danger of overambitiousness in the efforts of the totalizing philosophers, there may also be the danger of *underambitiousness* in the critics. There is a kind of smug safeness that very often goes along with the critical viewpoint. Many of these thinkers exhibit an uncreative tendency toward attacking the weaknesses in others while themselves lacking the sort of boldness required to build new systems of their own. It may be that these critical philosophers give up too easily and that they lack the courage and audaciousness that can be so inspiring in the totalizing visions of other thinkers.

Like the ends of a stretched rubber band, these two tendencies—the drive toward totality and the critical recognition of our failure to truly grasp totality—exist side by side and define the constant tension that underlies all real philosophy. Particular philosophers have been drawn to focus their attention on one or the other end of this “rubber band” at different times, yet a rubber band that is stretched too tightly snaps, and one that is left too slack stores no energy and falls limp. So it is with philosophy. If the drive for the Absolute is not tempered by the recognition of our intellectual limits, the field of philosophy “snaps” and relinquishes its questioning posture, giving in to the answers provided by science or religion. Conversely, if the margins of our intellectual boundaries are not constantly pushed by a passionate desire for the Absolute, philosophy “falls limp” and collapses into mere criticism and skepticism. At its most vital and active, philosophical thinking exists between these two poles. It is a tension that is never at rest with itself. Insofar as all of the thinkers we have surveyed in this book exhibit some degree of philosophical wisdom, even as they may be pulled to one extreme or another, this sort of tension is always present in their thinking.

David Hume, whose ideas we examined in the previous chapter, is an example of an individual who undertook one of the most ambitious and extreme attempts in history to force philosophy into falling limp. He thus typifies the radical end of the critical spectrum of the field. There are certain things that we can never know, Hume tells us, and so we should not speculate too vigorously on whatever it is that lies beyond the grasp of our knowledge. Cause and effect, the soul, and God in particular are things of which humans have always desired knowledge; however, because there is no possible empirical evidence that these things really exist, Hume tells us that we should just stop thinking about them. If we restrict our thoughts to those things that are within our grasp, Hume concludes, then we can live content and unperturbed by vain speculations. By talking only about those things that are empirically verifiable and in making ourselves useful to those around us, we can achieve happiness and reap the rewards that society has to offer us in the here and now. Of course, in counseling us not to worry too much about grand metaphysical issues, Hume himself gave

these issues a great deal of thought and thus, ironically, created his own form of metaphysical system. It was this system that would exert a massive influence on one of the most pivotal figures in modern philosophy, Immanuel Kant (1724–1804).

THE AWAKENING OF KANT

Hume's conclusions were disturbing to many people, and Kant was one of those people. Kant had been a dogmatic rationalist before encountering Hume's ideas, but he tells us that reading Hume "interrupted his dogmatic slumber."¹ Hume had conclusively shown, thought Kant, that the ideas of cause and effect, the soul, and God can never be found ready-made in the objective world. Because we never see, feel, taste, smell, or touch them, we have no empirical proof that they exist. Of particular concern to Kant was the idea of cause and effect. If cause and effect is an illusion, as Hume's philosophy implies, then we can never be certain whether or not the events we perceive in the world will consistently be followed by other events that we have come to expect. How do I know when I drop a ball off of a tower that it will fall downward and strike the ground? How do I know when I put my hand in a fire that it will be burned? How do I know that water freezes at zero degrees Celsius? If Hume is correct, then I don't know these things with certainty. I only think them probable because I have observed them to occur again and again in my past experience. But who says that the future may not be different from the past? If this is possible, furthermore, then what does this do to science, which is a field of inquiry that takes for granted the idea of cause and effect? Is science even possible after Hume?

It is this last question that propelled Kant as he began to write his most famous and influential work, *The Critique of Pure Reason*. This book was to be one of three critiques that would attempt to unearth the foundations of our thinking and to expose the necessary conditions that underlie our systems of knowledge, ethics, and aesthetic judgments. In the first critique, *The Critique of Pure Reason*, Kant addresses the topic of "what we can know" and attempts to delineate the outline of a science of human knowledge. In the second critique, *The Critique of Practical Reason*, Kant addresses the topic of "what we ought to do," attempting to demonstrate



the basis of all human morality. In the third critique, *The Critique of Judgment*, Kant addresses aesthetics and attempts to show what it is that makes our judgments of beauty and sublimity possible. Taken together, the three critiques represent an ambitious attempt to comprehensively examine the structures of human thinking and to show that while our knowledge does have limits, those limits are precisely what make human forms of science, morality, and aesthetics possible. Kant's work, in this way, truly embodies the tension that is a unique part of philosophical thinking. On the one hand, he will argue that the human mind is incapable of knowing everything that there is to know about objective reality. On the other hand, he will argue that our subjective reality is shaped not just by the inborn structures of the mind but also by the human desire to transcend the pregiven boundaries of our thinking. Ultimately, the picture that Kant paints is one in which humans constantly strive, yet fail, to achieve an understanding of the Absolute. Despite our failures, we nevertheless do achieve progress in our thinking through our constant aspiration toward the unattainable. Thus, while Kant's philosophy is critical of those who claim to have perfected a final understanding of absolute reality, he at the same time grants that the drive toward totality has an important purpose in human thinking and should for this reason not be discouraged.

Kant lived a life that was by all accounts very structured, regular, and conventional. Otfried Höffe writes, "Nothing extravagant taints his life-style: no unusual clothing or hair style, no moving gesture such as the 'Sturm und Drang' period loved."² Kant was, it appears, not a very dramatic or exciting figure in terms of his day-to-day activities. Though he was a well-loved teacher, he never experienced any grand adventures, and in fact during his lifetime he never even left the city of Königsberg where he was born. It is claimed that his neighbors would set their watches according to his daily routines, so regular were his habits. This sort of stable and predictable way of life may have been a reflection of his underlying desire for orderliness and structure, which would in turn contribute to his later attempts to systematize all human thinking in his three *Critiques*.

Over the course of his early writing career, before his "critical period," Kant was fascinated with science and astronomy, publishing works that speculated on the origins of the universe and earthquakes. Philosophically, he was at first a dogmatist who believed that our understanding of the world proceeded by the analysis of ready-made concepts and their connections to one another. During this early "precritical" period, the philosophical works that he produced were sometimes tedious and uninspiring. One prime example of this is his essay *Observations on the Feeling of the Beautiful and Sublime*,³ a work that simply regurgitates uncritically a variety of old stereotypes about men, women, and the world:

Women have a strong inborn feeling for all that is beautiful, elegant, and decorated. Even in childhood they like to be dressed up, and take pleasure when they are adorned. They are cleanly and very delicate in respect to all that provokes disgust. They love pleasantries and can be entertained by trivialities if only these are merry and laughing. Very early they have a modest manner about themselves, know how to give

themselves a fine demeanor and be self-possessed—and this at an age when our well-bred male youth is still unruly, clumsy, and confused.⁴

These sorts of generalizations today strike us as silly and unwarranted, and if Kant had never broken free from this sort of dogmatic and conventional way of regarding the world he would not have become such a pivotal figure in the development of Western thinking. To his credit, and in contrast to his otherwise conventional, comfortable, and predictable lifestyle, Kant was not unwilling to take chances and break new ground intellectually. It was only upon emerging from his early precritical period that Kant was to become one of history's most important philosophers.

As mentioned above, Kant credits Hume for rousing him out of his “dogmatic slumber.” Upon reading Hume's works, Kant realized that there was no possible way that universal laws and rules, such as the law of cause and effect, could be deduced from observation. Does this mean then, as Hume had concluded, that ideas like cause and effect are nonsense? According to Kant, the answer is “no.” Certainly Hume was correct that our idea of cause and effect is not derived from experience, but this does not show that the idea is gibberish, only that it must be derived from something other than experience. Kant points out that the idea of cause and effect is entirely indispensable for the scientific knowledge that we do claim to have about our world. Without cause and effect we would live in a weird world that is unpredictable and strange; a world in which it would be unsurprising to find that balls dropped from towers fall upward, or that hands placed into fires become cold, or that water freezes at 1,000 degrees Celsius. This is not the world that we live in, however. The human world of experience is the world that makes sense to us, and it is one that takes for granted the relationship of cause and effect. Hume's mistake, Kant would claim, was that he looked for the origin of causality in the wrong place. Cause and effect does not exist somewhere “out there,” independent of human experience. It is, rather, one of the preconditions for human experience. It is not found in any of the impressions we have of the world but is imposed by our minds upon the ideas that arise from those impressions. According to Kant, Hume's criticism of the idea of cause and effect is misguided because Hume forgot that like the principles of resemblance and contiguity in space and time, cause and effect is an important organizing principle of the mind itself. Without the imposition of some form of organization upon sensory input, the world would appear like a chaotic mass of unconnected and disassociated ideas. Because this is not the way that the world does in fact appear to us, our minds must indeed impose some kind of order on ideas. Cause and effect is just one of those principles of order.

THE CRITIQUE OF PURE REASON

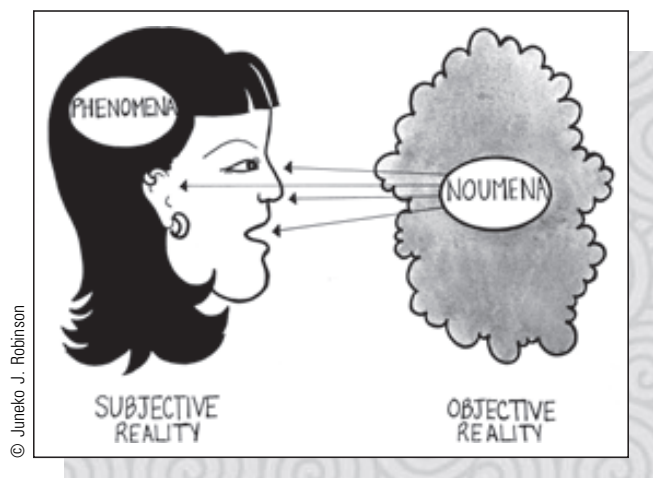
At the very beginning of *The Critique of Pure Reason*, Kant makes the keen observation, “... though all our knowledge begins with experience, it does not follow that it all arises out of experience.”⁵ With this, he disputes the basic assumption that empiricists like Hume take for granted. We cannot account for the way the world appears to us if we presume, as Hume does, that our physical

senses are the only guide that we have to reality. Instead, it must be the case that in addition to sensory experience, the mind itself contributes certain modes of organization that it imposes on the data of perception. Knowledge, then, is not simply the collection of sensory impressions, but rather the collection *and* the organization of sensory impressions. In other words, to really *know* something our minds must both take in information and process that information to make sense of it. Knowledge is constituted in this way and is the product of an active mind that sifts through data and systematizes it according to a schema of conceptual categories.

The Phenomenal and Noumenal Worlds

Kant suggests that we shift our attention away from speculation about the constitution of the objective world and instead focus our attention on the world as we subjectively experience it, asking the question, "What makes that world of subjective experience possible?"⁶ Think about how different creatures experience their surroundings. Bats, for instance, make sense of their surroundings predominantly by using something called "echolocation." They send out sound waves that bounce off of objects and return to their auditory sensors with information that helps them to determine where things like trees, rocks, and other creatures are. Their world is built up out of information that is mostly auditory. This is a much different world from that of an earthworm, which has no auditory sensors. The earthworm's world is a world of touch. It feels the earth's soil on its body and navigates accordingly. Likewise, there are many other creatures that build worlds around them by relying upon one or more of the other senses. The manner in which they experience things differs according to how they encounter and impose structure upon the sensations that are created in them by way of interaction with the outside environment.

While there is presumably a world "out there" that exists independently of experience, no creature can know that world independently of experience. All creatures, humans included, must process their impressions of the world through a perspective, and while that perspective gives a hint of the world beyond experience, it also distorts and hides certain aspects of reality. Kant calls the world as it is perceived the "phenomenal world." The phenomenal world is the world of experience that emerges from the interaction between the mind and the reality that exists independently of the mind. That which exists independently of the mind Kant calls the "noumenal world."⁷ It is the world of uninterpreted reality that would be there regardless of whether or not we encountered it. The phenomenal world is the only world that we have direct experience with, while the noumenal world is the "thing-in-itself" beyond our direct powers of perception. We only encounter the noumenal world by way of filtering it through the lens of the mind. But as we filter our perceptions in this fashion, the phenomenal world emerges, and thus any time that the mind encounters reality, the reality that it encounters is the world of subjectively interpreted phenomena and not the objective world as it exists in-itself. Think about the way that we perceive tastes or colors, and you will get a good idea of what Kant is driving at here. Tastes and colors are phenomena that exist wholly within the mind of the perceiver. They do not exist in the objective world itself as they are perceived,



but rather are produced through the interaction of human sensory mechanisms with something from the objective world. The interaction between a sensory organ, say the eye, and something in the environment, say a wavelength of light, produces in the human mind the perception of a color, say red. As Kant writes, “The taste of wine does not belong to the objec-

tive determinations of the wine ... but to the special constitution of sense in the subject that tastes it. Colours are not properties of the bodies to the intuition of which they are attached, but only modifications of the sense of sight, which is affected in a certain manner by light.”⁸ And so it is for all of our encounters with reality according to Kant. Our minds are constantly interpreting and processing information, and because we can never step outside of our own minds, all that we know of the world is the interpretation that constitutes our phenomenal reality.

Who has a better idea of what the world is really like: the bat or the earthworm? In a way, this is a silly question. The bat and the earthworm live in subjectively different worlds. The earthworm knows its world, and the bat knows its own world. They have differing phenomenal experiences by which they conceptualize reality. Certainly, their worlds may overlap at points, such as when the earthworm feels itself being eaten by a bat that has hunted it down by means of echolocation. But to say that the phenomenal world of the bat more accurately represents objective reality than does the phenomenal world of the earthworm misses the point that neither of their phenomenal worlds corresponds precisely with the noumenal world, or the world as it is in and of itself. Both creatures must filter through and organize the sensations they receive through their sensory organs to know any world at all. But the world that they come to know is not necessarily anything like that which exists apart from their perceptions of it. Any knowledge that a mind has of the world “has only to do with appearances, and must leave the thing in itself as indeed real *per se*, but not known by us.”⁹ Thus, whatever the noumenal world consists of, there is no creature, no matter what its perspective, that can possibly know its true and objective nature. Because knowledge consists in the interpretation and organization of sensory data by the mind, all knowledge is locked into one perspective or another. There is no such thing as a mind that knows uninterpreted reality.

The *a Priori* Intuitions of Time and Space

So what is it that makes the *human* perspective distinctive? Like any conscious creature, humans must interpret the sensations drawn from their environments,

and we do so, claims Kant, by means of organizing sensory input according to a unique schema that already exists inside our minds. While our mode of perception is “not necessarily shared in by every being,”¹⁰ the first stage of every *human* being’s perception involves what Kant calls the “*a priori* intuitions of space and time.” All humans, according to Kant, understand the world as unfolding within the domain of space and time, and yet because space and time are not sensations that we can perceive with our senses, our awareness of them must be derived from something other than empirical, sensory data. Space and time, in fact, must originate *a priori*, or before experience, because they are the preconditions for all of our experiences. Therefore, space and time do not exist outside of us in the objective, noumenal world but rather are intuitions of the mind itself that are imposed upon the data of experience as the first mode of phenomenal organization. The human, phenomenal world always occurs within the constraints of space and time, claims Kant, precisely because the human mind itself necessarily imposes this structure on all of its experiences.

Space is the precondition for the mental representation of outer experience, while time is the precondition for the mental representation of inner experience. What Kant means by this is that to think about objects existing in the world outside of us, we must first presuppose open space in which those objects may appear.¹¹ Though we never actually see space, our minds must conceive it to distinguish distinct objects existing independently of one another and separate from us. If there was no space, there would be no place for our representations of objects to unfold. Everything would be one thing; or more accurately, everything would be nothing, because there would be no things! Likewise, time is something that we never empirically observe; however, to be aware of our inner world of representation we must think thoughts in temporal order, one after the other. If our minds did not impose this sort of sequential ordering on our representations, then everything would appear to us to occur all at once. Furthermore, it is only because of the ability to organize representations according to a “time sequence” that we are able to become aware of our own inner state of consciousness. The power to think things in sequence attunes us to the world of inner thought, which is a world that we never empirically observe but only experience intuitively during the temporal process that is thinking.

Kant’s analysis of the intuitions of space and time explains how it is that the pure sciences of geometry and mathematics are possible. A pure science is a science that is completely *a priori* and not based on empirical observation. According to Hume’s metaphysical system, pure sciences are not possible, because all ideas must be traced back to sensory impressions to be meaningful. Kant, however, claims that our knowledge of geometry is completely based on our inborn intuition of spatial relationships as they appear ready-made in the mind. Likewise with mathematics. Pure mathematics is possible as a science because we are endowed with an inborn intuition of time, which makes possible the temporal ordering of particulars, one after the other, enabling us to count, as in arithmetic. We need not observe anything at all to learn these sciences; we are already hard-wired for them. Furthermore, and contrary to Hume, Kant asserts that these pure sciences are informative. Through their productive synthesis, the *a priori* intuitions give us new information and not just the completely empty formalism

that Hume had claimed was the earmark of any analytic system. When we add 1,023,000 to 2,800,000,000 and then divide the sum by 6.5, we learn something new, according to Kant. We gain new information from this *a priori* synthesis that was not apparent beforehand. According to Hume, such a procedure only teases out information that was already there, and so nothing new is learned; it is merely a “relation of ideas.” For Kant, however, there is such a thing as *a priori* synthetic knowledge; or in other words, knowledge that is completely free from experience but also meaningful and informative.

With his account of the *a priori* intuitions of time and space, and with his claim that these intuitions are sufficient to ground the pure sciences of geometry and mathematics, Kant believes that he has undermined the empirical presumptions of Hume. It is simply not the case, Kant concludes, that all meaningful knowledge comes from observation and experience. Rather, the mind possesses within itself a set of structures that work in concert with one another, producing knowledge and information that is absolutely certain, and that acts as the ground for the nonobservational sciences. Now that he has discovered the foundation for these nonobservational sciences, Kant considers what it is that grounds the observational sciences.

The Categories of the Understanding

While geometry and math deal with abstract entities that are completely generated from within the mind, the observational sciences, like physics and astronomy, purport to deal with objectively real entities that exist outside of our minds. But if these objects exist outside of our minds, how can we be certain about the manner in which they will always behave? Hume had tried to convince us that because the future can always be different from the past, there is no possible way of predicting with certainty how the world of natural objects will operate into the future. Kant, however, is not satisfied with this. He points out that we do, in fact, have a science of nature that seems to work well, as is apparent in fields like physics and astronomy. So instead of skeptically dismissing these accomplishments, Kant instead seeks to inquire into the conditions that make these sciences possible. “Since these sciences actually exist, it is quite proper to ask how they are possible; for that they must be possible is proved by the fact that they exist.”¹² So how is it possible that we have a science of nature?

Natural science is the study of the relationships between natural objects, and as such it presupposes not only a field of time and space within which these objects may appear but also a logic by which the connections between them may be known. Unlike the pure sciences, the natural sciences utilize empirical input that the mind collects from its sensory mechanisms. While the mathematician need only think about numbers, the astronomer must actually look to the skies and the physicist must actually make measurements of the physical world. Thus, the natural sciences utilize *a posteriori* observation in addition to *a priori* forms of mental organization. This is what is distinctive about the science of nature, then. It brings the mind together with sensation in order to build up a picture of the world around us. Without sensory input, our mind would have nothing to work with. Without innate organizational structures, sensory input would have no

logic or order. As Kant writes, “Without sensibility no object would be given to us, without understanding no object would be thought. Thoughts without content are empty, intuitions without concepts are blind.”¹³ In order to have a science of nature you must have both an object of sensation and a way of thinking about that object of sensation in connection with other objects of sensation. It is clear that particular objects of sensation are given to us by way of the mind's space- and time-bound perceptions, but what is the pattern of logic by which we connect these perceptions together into a coherent and orderly whole?

Kant finds an answer to this last question in the philosophy of Aristotle. Recall that Aristotle was the first person to offer a systematic account of logic, which he claimed was the basis of all human thinking. Kant picks up on this and applies Aristotle's logic to an analysis of the very structure of human understanding. The mind, Kant tells us, not only intuits objects within the context of space and time, it also has in place a system of twelve categories by which it understands the connections between these objects. These twelve “categories of the understanding” represent an exhaustive list of the ways that perceptions may be synthesized within the human mind, and they are enumerated by Kant as in Table 9.1.

TABLE 9.1 Kant's Categories of the Understanding¹⁴

I. Of Quantity Unity Plurality Totality	II. Of Quality Reality Negation Limitation
III. Of Relation Of Inherence in Substance Of Causality and Dependence Of Community	IV. Of Modality Possibility–Impossibility Existence–Nonexistence Necessity–Contingency

What is interesting to notice in this list of the categories is how Kant has completely subjectivized a number of relationships that had previously been assumed to exist outside of the human mind in the objective world of nature. Take, for instance, the first group: “Of Quantity.” Here we have the categories of unity, plurality, and totality listed as organizing structures of our thinking. What Kant is telling us is that when we look at the world and process the information that we receive from that world, it is the mind, and not the world in-itself, that is organizing what we perceive according to quantity. Quantity, in other words, does not exist “out there” independent of us, but is rather a way for us to make sense of our environment. There is no such thing as “unity” or “plurality” or “totality” out there in the objective, noumenal world. Rather, we sift through and interpret whatever it is that is “out there” and make sense of it in these terms. Physicists today tell us something quite similar to this when they point out that what appear to us as solid objects in our environment are really not solid at all but mostly made up of empty space. A table, for instance, is made up of subatomic particles that have more space between them than they

have firm matter. Why is it, then, that a table appears solid to us? Because our minds have a tendency to group particular particles together and to see them as unified objects. Experiments have shown, in fact, that when objects are multiplied beyond a certain number, test subjects begin to have problems distinguishing between individual members of the group and so begin to see them as a unified whole. Consider what happens when you gaze out at a vast number of trees. Instead of seeing individual trees you are likely instead to see a forest. This is just the difference between interpreting what you see as a plurality or a unity, and it is all going on inside your mind. The trees themselves stay the same. It is only your way of thinking about them that changes.

Just as the categories that define “quantity” are contributions from the mind, so are the categories that define “relation.” Subsumed under this third grouping are particular classes of relationships like “substance” and “causality.” These two latter categories are especially interesting given that they are the very things that Descartes claimed to discover by meditating on the structure of his own thinking and that Hume claimed were absent from his sensory impressions of the world itself.¹⁵ According to Kant’s analysis, neither substance nor causality exist independent of human thinking; rather, they are actually part of the mind’s own organizational structure, just like all of the other categories of the understanding. A “substance” is an underlying and unchanging foundation that supports the sorts of properties that cannot exist in and of themselves. For instance, Descartes concluded that mind substance must exist because thoughts, which cannot exist without someone to think them, obviously exist. Yet one cannot empirically observe mental substance, and for this reason Hume concluded that Descartes was wrong about its existence. Kant, on the other hand, makes the claim that mental substance does exist, but only as an organizing strategy that allows us to impart coherence to impressions that would otherwise seem quite fleeting and unpredictable. For instance, if we didn’t have the idea of a mental substance, all we would be aware of would be transitory thoughts that pop into and out of consciousness. By activating the category of substance, however, we bring all of our thoughts together, in a sense, and ground them by rooting them in a common underlying concept that persists through time even as the thoughts themselves change.

Cause and effect is another idea that Descartes claimed was clearly and distinctly real, while its existence was disputed by Hume. Recall that cause and effect involves the necessary connection between two or more ideas. We say that the appearance of one thing is a cause if it always brings about the appearance of another thing, the effect. Hume rejected the idea of causation because it is impossible to empirically observe a necessary connection between two or more things. According to Hume, all we can observe is a relationship of constant conjunction by which ideas repeatedly accompany one another, but we can never be sure that they will always accompany one another because we cannot see into the future. Kant, however, asserts that while Hume is correct that evidence of causation is not to be found in our observations of the world, it is nonetheless one of the ways that we make sense of our observations of the world. It is one way that the mind imparts structure to ideas. Of Hume, Kant writes, “... it never occurred to him that the understanding might itself, perhaps, through these concepts, be the author of the experience.”¹⁶ Thus, Kant puts a new slant

Box 9.1 Autism and Kantian Philosophy

Autism is a mental disorder in which people are unable to interact normally with others. Symptoms become apparent within the first three years of life. Typically, autistics seem withdrawn and internally preoccupied. They engage in behaviors like rocking, hitting themselves, and various other forms of self-stimulation.

In the past it was thought that autistics were completely unable to function in the normal world. However, in recent years there have been a number of people, previously labeled as autistic, who have come out of their shells, not only describing their experience but also becoming professionally successful.

Temple Grandin was diagnosed with autism, yet she has earned a PhD in animal science and has become one of the most successful designers of livestock facilities in the United States. In her book *Emergence*, Dr. Grandin describes the experience of being autistic in terms that recall Kant's description of the human mind. She reports that in autism, one is not capable of properly filtering the stimuli that come through the senses. As a result, the autistic person becomes overwhelmed by input from the outside world and so withdraws.

Grandin's account matches that of Donna Williams, who in her book *Nobody Nowhere* also discusses the subjective experience of being autistic. Like Grandin, Williams describes being unable to filter out the chaos of the objective worlds stimuli. Williams has now published ten books while also painting and working on a screenplay.

Autism might be thought of as a breakdown of the categories of the understanding. Those who are not able to mentally organize experience are perhaps apt to withdraw from the overwhelming chaos of the noumenal world.

on Hume's own assertion that the mind has certain "habits" when it comes to the association of ideas. But while Hume dismissed the idea of cause and effect as a "bad habit," Kant upholds it as a necessary mechanism for the organization of our thoughts.

In combination with one another, the *a priori* intuitions of space and time and the twelve categories of the understanding give order and coherence to our phenomenal experiences. These experiences, it needs to be reiterated, are not an exact mirror of the noumenal world that lies behind them but are rather interpretations that together constitute our shared human version of reality. While the human, phenomenal world may be different from that of a bat or an earthworm, it is nonetheless our reality. Time, space, substance, causality; all of these things really do exist for us, although they do not exist independent of us. As our minds encounter sensory stimuli, they organize and synthesize these stimuli, constructing a pattern of phenomenal appearances that constitutes the world we are conscious of and within which we make our lives.

The picture that Kant gives us of the human mind characterizes it as something like a very sophisticated computer program. It has in place a whole set of formal structures that await the input of content so that it can begin to categorize data and organize them into useful information. Consider the way that a web browser functions. If you were to look directly at the source codes constituting the raw data that web browser programs read, they might look like a bewildering array of nonsense. However, when you run this seeming nonsense through the

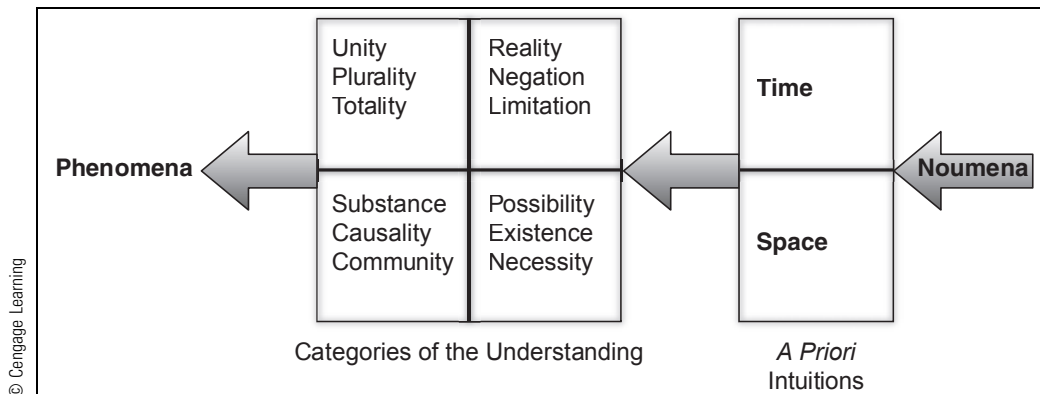


FIGURE 9.1 The Constitution of Phenomenal Reality

web browser program, the program makes sense of the data, interpreting them in terms of a well-organized screen of information. Comprehensible pictures, text, and sounds emerge through the organizational powers of the web browser. Likewise, the human mind, according to Kant, possesses certain innate structures that allow us to give significance to the data reaching us through the senses. Our phenomenal world emerges in our minds like images, words, and sounds emerge onto a computer screen. By organizing data according to the *a priori* intuitions and the categories of understanding, a comprehensible world-picture emerges, and though it may bear only as much resemblance to the objective world as the image on a web browser bears to its source code, it is this emergent world that we call our own (see Figure 9.1).

Transcendental Idealism and the Impossibility of Metaphysics

Kant's philosophy is called "Transcendental Idealism" because it attempts to transcend the ideas in our minds in order to get at the conditions that make those ideas possible. Transcendental Idealism can be thought of as an attempt to synthesize the insights of both rationalism and empiricism. On the one hand, Kant agrees with the rationalists that there are certain innate structures of the mind that provide grounding for our knowledge of the world. On the other hand, he agrees with the empiricists that without sensory input, these innate structures cannot do their job. Without experience, our minds would be like empty file cabinets or computer programs without data input. So it is that Kant walks a middle ground between the extremes of Descartes and Hume.

Though his philosophy seems very sensible and moderate in a number of ways, there is at least one troubling consequence to be drawn from Kant's characterization of human knowledge. If Kant is right, then what we call "knowledge" is not knowledge of the world as it objectively exists. We can never know with certainty what the world is like independent of our experience because all we can experience and know with certainty are the phenomena that arise out of an interaction between our minds and the noumenal world. We always experience the noumenal world only through the subjective lens of our minds, which color and distort reality as it objectively exists independent of us. Whatever objective

reality is like, we cannot know its essence. If this is true, then all of our knowledge is only subjectively valid. If Kant is correct, then truly objective knowledge is impossible. We must forever be separated from the world as it is in-itself by our uniquely human perspective. Locked into the human perspective, we can never know what lies beyond that perspective. We are like prisoners in a cave.

Kant's Transcendental Idealism was intended as a means of rescuing our knowledge from the sort of skeptical attacks that Hume had undertaken. We do have absolutely certain knowledge, claims Kant, but this absolutely certain knowledge is not of the noumenal world. Rather, it is knowledge of the phenomenal world of human experience. All knowledge is subjectively constituted, according to Kant, and for this reason, while a science of phenomena is possible, a science of metaphysics is not. Metaphysics is the science of being-in-itself, but because we never have experience of being-in-itself, any attempt to argue about its nature is doomed to failure. Because our human perspective cuts us off from the world as it is in-itself, the only science that is possible for us is a science of appearances.¹⁷ Now, this has not stopped past philosophers from arguing endlessly about things like the nature of the soul, the beginning of the universe, and the existence of God, but according to Kant such arguments must always fail precisely because there is no observational evidence that can ever be produced in order to settle these issues. The problem with philosophical debates of this sort, Kant tells us, is that they rely on "pure reason." When pure reason is exerted without sensory input, the human mind is capable of producing all sorts of competing and inconsistent arguments that have the appearance of being logical but which always leave us with some sort of unresolved problem. For instance, you can offer formally valid arguments concluding that the universe has a beginning, but you can also offer formally valid arguments that the universe is eternal, says Kant. These arguments on their own will never settle the issue at hand because they attempt to derive conclusions about things that are not within the realm of potential experience. When you have good arguments on either side of an issue, the only way to figure out which argument is correct is to test the arguments against observation. However, because metaphysical arguments concern the noumenal world, and we cannot possibly ever experience the noumenal world, these arguments must remain unresolved.

Metaphysical arguments are examples of what Kant calls "transcendental illusions" produced by the exercise of pure reason. At the same time that the various categories of the understanding serve to sort through and coherently unify our sensations, the faculty of pure reason "secures the unity of the rules of understanding under principles."¹⁸ Pure reason, in other words, works toward the completion of the task that the understanding only begins, which is the absolute unification of all thinking under a supreme and unconditioned principle. It is this tendency toward the absolute unification of thinking that is the source of the "transcendental illusion" as it is found in arguments about the soul, the origins of the universe, and God's existence. Pure reason desires a final end to the series of conditions that the mind sets in place over the course of attempting to synthesize an understanding of the world, and for this reason it has a tendency to reach beyond its grasp, trying to come to conclusions about issues that it cannot

possibly establish through the function of the understanding alone. In the course of this aspiration, pure reason produces “transcendental ideas,” or principles that go beyond all possible experience, but which also serve to focus and consolidate the manifold variety of all the other ideas we have in the course of our thinking lives. As we saw with St. Aquinas’ arguments for God’s existence in Chapter 6, for example, the transcendental idea of God serves as a final anchor point in which all else is rooted. Everything culminates in God, and so the idea serves the function of giving pure reason a sense of completion and satisfaction. All of the loose ends get tied together, and thinking can come to an end. Pure reason, Kant claims, always strives toward this sort of culmination.

The Regulative Function of Transcendental Ideas

Though the exercise of pure reason results in transcendental illusion, according to Kant this is not such a bad thing. In fact, it is only natural and in some ways it is downright useful. While “all those conclusions of ours which profess to lead us beyond the field of possible experience are deceptive and without foundation,” it is also the case that “human reason has a natural tendency to transgress these limits.”¹⁹ This natural tendency of reason to go beyond all possible experience and to strive toward the unification of all knowledge serves to give thinking an ultimate purpose, directing it toward a goal that, while quite unattainable, is nevertheless useful in its “regulative” function. If our thinking was always pulled this way and that way without any focal point, progress would never be possible in any area of our intellectual lives. The transcendental ideas, while illusory, serve the purpose of beckoning us in a particular direction and of motivating us to develop and hone our way of thinking so that it becomes progressively more unified, concentrated, and organized (see Figure 9.2). As Kant writes:

[Transcendental ideas]
... have an excellent,
and indeed indispensably
necessary, regulative
employment, namely
that of directing under-
standing towards a cer-
tain goal upon which
the routes marked out
by all its rules converge,
as upon a point of in-
tersection. The point is
indeed a mere idea, a
focus imaginarius, from
which, since it lies quite
outside the bounds of

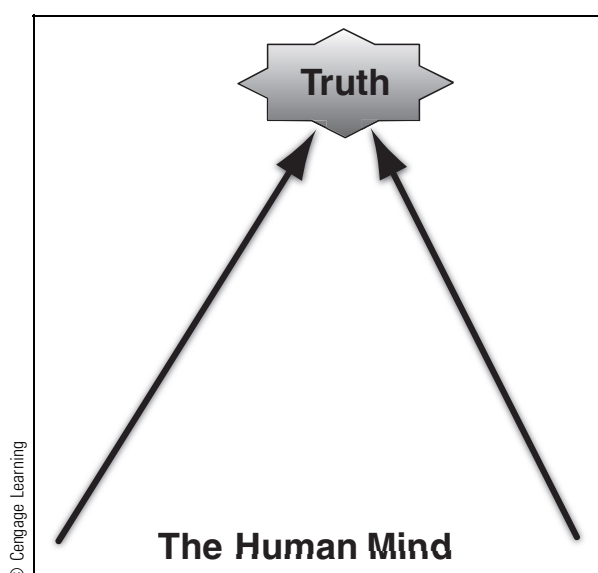


FIGURE 9.2 The Mind’s Aspiration Toward the Unification of Knowledge

possible experience, the concepts of the understanding do not in reality proceed; nonetheless it serves to give these concepts the greatest unity combined with the greatest extension. Hence arises the illusion that the lines have their source in a real object lying outside the field of empirically possible knowledge—just as objects reflected in a mirror are seen as behind it. Nevertheless this illusion ... is indispensably necessary if we are to direct the understanding beyond every given experience ... and thereby to secure its greatest possible extension, just as, in the case of mirror-vision, the illusion involved is indispensably necessary if besides the objects which lie before our eyes, we are also to see those which lie at a distance behind our back.²⁰

THE CRITIQUE OF PRACTICAL REASON

Human reason, Kant claims, is always trying to overreach its own boundaries. As it pushes toward greater and greater degrees of theoretical unification, reason will never rest until it has formulated a single principle under which all knowledge can be subsumed. While this task can never be completed, the human mind naturally desires its completion and thus endlessly aspires toward a theoretical principle that is beyond its grasp. But in addition to this *theoretical* application, reason also has a *practical* application. Just as in the theoretical realm reason seeks the unification of all knowledge under a single principle, in the practical realm it seeks the subsumption of all actions under a universal action-directing principle. This is the task of practical reason, which involves the use of the faculty of reason for the formulation of universal laws delineating how we ought to act in the world. Whereas pure reason seeks an absolute unity of knowledge, practical reason seeks an absolute principle of morality.

In the preface to *The Critique of Practical Reason*, Kant notes that there was a certain “critic” who once complained that Kant’s moral writings “contain no new principle of morality.”²¹ Kant’s amused response to this criticism is to joke, “Who would want to introduce a new principle of morality and, as it were, be its inventor, as if the world had hitherto been ignorant of what duty is or had been thoroughly wrong about it?”²² The point that Kant is driving at with this humorous remark is one with which thinkers like Socrates and Plato would also agree. It would be ridiculous and misguided to consciously contrive a new principle of “morality” because that which is truly moral is so independent of what people say or think. You cannot just make up new moral principles. The purpose of reasoning about morality and ethics is not to come up with novel, clever, and unprecedented rules but to clarify and elucidate the guidelines of proper action that our intuition tells us are true already. In our world, people sometimes act morally and sometimes they act immorally. We make these sorts of determinations every day. How is it that we do so? The task of practical reason is to answer this question. Just as Kant, in *The Critique of Pure Reason*, purported to uncover the *a priori* conditions that make science possible, so too in *The Critique of Practical Reason* does he attempt to demonstrate the underlying *a priori* conditions that make morality possible.

KANT'S WONDROUS DISTRESS

“Two things fill the mind with ever new and increasing wonder and awe, the oftener and the more steadily we reflect on them: the starry heavens above me and the moral law within me.”³² In this, one of the most famous quotes from Kant, is summed up his sublime respect and reverence for both the inner and the outer worlds of human experience. Kant began his writing career distressed by a crisis in human knowledge, and his systematic critique of human thinking represents a sustained attempt to demonstrate the universal and necessary *a priori* principles that our minds rely on to organize, synthesize, and interpret the sensations we receive from the objective world.

Kant, however, did not end up offering us final answers to all of the mysteries of existence. At the heart of the human soul, and in the depths of the outer universe, there remain mysteries that we can never fully understand. While our innate drive for unlimited knowledge pushes us forward, our separation from the objective world by way of the distinctively human perspective will continue to foil our plans for a final apprehension of the absolute. Kant in this way was both a systematizer and a critic. In offering a systematic account of how human thinking works, he also offered a criticism of those philosophers who claimed to have transcended the boundaries of all possible experiences. Kant tells us that he has “found it necessary to deny knowledge, in order to make room for faith.”³³ He denies absolute knowledge of the objective, noumenal realm, while granting us knowledge within the purely human, phenomenal realm. Beyond the phenomenal realm one may have faith in what one wishes, but such claims can never be substantiated by reason.

In these ways, Kant clearly exhibits the elements of wondrous distress that characterize truly philosophical thinking. In genuine wonder and awe, Kant strains to pursue ideas to their limits, only to find in the end that these limits constantly recede into the distance, the way that parallel lines disappear into a vanishing point and never converge. The distressing awareness that absolute knowledge constantly eludes our grasp, however, is presented by Kant as an opportunity for endless investigation of the world rather than as an excuse to give up on learning. The innate tendency of the human mind to strive toward complete and final knowledge of reality, though doomed to failure, can be harnessed in the service of ongoing and productive advancement in the human sciences. This is typified by the regulative power of transcendental ideas, which while technically presenting us with an illusory hope for completeness give us the practical motivation to act in the world. As we strive to encompass more and more knowledge under a more and more comprehensive and unified set of conceptual principles, our own human interpretation of reality progresses to become more panoramic, well textured, and sophisticated. Though the phenomenal world of human experience will always remain alienated from the noumenal world as it is in-itself, it is precisely this alienation that promotes our collective movement forward toward the ideal of Truth itself.

Kant's philosophical system promoted both inspiration and despair in those who came after him. As we will see in the following chapters, wonder and distress would be renewed in a whole new generation of philosophers as a result of Kant's work.